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SCHEDULE: Standards

IT is hereby notified that the Minister of Energy and Power Development, after consultation with the Zimbabwe Energy Regulatory Authority, in terms of section 65 of the Electricity Act [*Chapter 13:19*], has made the following regulations:—

Title

1. These regulations may be cited as the Electricity (Electric Vehicle Charging Station Safety) Regulations, 2026.

Interpretation

2. In these regulations—

"alternating current" means a type of electrical current in which the direction of the flow of electrons switches back and forth at regular intervals or cycles;

"consignment based conformity assessment" means a process for ensuring that a consignment to be shipped into the Zimbabwean market complies with national standards, relevant international standards, and technical regulations;

"direct current" means an electric current that is unidirectional the flow of charge is always in the same direction;

"electric vehicle charging station" refers to a power supply device that supplies electrical power, be it alternating current or direct current, for recharging plug-in electric vehicles;

"electric vehicle" means a vehicle that uses one or more electric motors for propulsion;

"electric vehicle supply equipment" means any equipment or electrical component used in charging electric vehicles at a specific location. Electric vehicle supply equipment does not include equipment located on the electric vehicles themselves (on board chargers);

"International Electrotechnical Commission" refers to the international standards and conformity assessment body for all fields of electro-technology;

"minimum energy performance standards" specify the minimum level of energy performance that electric vehicle charging stations must meet or exceed before they can be imported, distributed, or offered for sale or use in Zimbabwe;

"open charge point protocol" means an application protocol for communication between electric vehicle charging stations and a central management system also known as a charging station network;

"residual current device" means a device that is able to detect a fault current also in the absence of neutral;

"TN-C-S (MEN)" means an earthing system arrangement for low voltage network that protects people and structures from electrical hazards;

"ZWS" means the Zimbabwe Standard.

Application

3. These regulations shall apply to all importers and retailers, contractors, installers, and owners of electric vehicle (EV) charging stations.

Objects

4. The objectives of these regulations are for the promotion of electrical safety in the installation and usage of EV charging stations.

Compliance with laws and standards.

5. (1) All EV charging stations must be designed, installed, tested, certified, inspected, and connected in such a way as to ensure the safety of the users and the public in accordance with the Electricity (Public Safety) Regulations, 2018, published in Statutory Instrument 177 of 2018, SAZ ZWS400, SAZ ZWS913:2010.

(2) All EV charging stations must—

(a) be supplied from a dedicated final sub-circuit; and

(b) be supplied from a sub-circuit protected by a voltage-independent residual current device (RCD) providing personal protection that is compatible with a charging supply for an electric vehicle; and

(c) provide an earth continuity monitoring system that disconnects the supply in the event that the earthing connection to the vehicle becomes ineffective. This monitoring system must fail to safety; and

(d) provide protection against the overload of the charging supply fittings; and

(e) provide protection against the overload of the incoming supply fittings; and

(f) be supplied from a TNC-S (MEN) system of supply that includes an earth electrode and MEN connection; and

(g) be installed so that any socket-outlet of a Mode 3 or Mode 4 supply is at least 800 mm above the finished ground level; and

(h) for charging stations with multiple connecting points, all connecting points must be protected by a voltage-independent RCD that is compatible with a charging supply for an electric vehicle; and

(i) all RCDs for the protection of connecting points and electric vehicle charging stations must incorporate the ability to continue to protect in the event of above 6mA of direct current (DC) fault, including leakage, current or isolate the supply in the event of more than a mA of DC fault, including leakage, current. Paragraphs (c) and (g) do not apply to wireless charging stations.

(3) No EV charging station may provide or be connected to an alternating current (AC) charging supply that—

(a) is not at standard low voltage; and

(b) is not at a nominal frequency of 50 Hertz; and

(c) all functions of an EV charging station must fail to safety.

(4) Any person offering for sale or supplying an EV charging station must have, and must make available to the Authority, purchaser, potential purchaser or installer a supplier declaration of conformity—

(a) in respect of an EV charging station complying with IEC standards, the relevant test reports, approvals or other document as specified by the Authority; or

(b) in the case of an EV charging station complying with UL standards, the relevant UL certification verification, or other document as specified by the Authority.

(5) Any person installing an EV charging station must verify, keep, and attach to the Certificate of Compliance records that the EV charging station has been designed, constructed and labelled to be compatible with a supply of standard low voltage at a nominal frequency of 50 Hertz.

(6) Any person who contravenes this section shall be guilty of an offence and liable to a fine not exceeding level five or to imprisonment for a period not exceeding six months or to both such fine and such imprisonment.

Electric vehicle charging station setup

6. (1) Any EV charging station and connected support devices shall be designed and installed to operate on the Zimbabwe electricity distribution network and take into consideration the distribution code's power quality requirements and installation standard as provided in the Schedule.

(2) EV charging stations shall include mandated settings that automatically operate to protect both the customer's electrical installation and the distribution network it is connected to, in a bid to ensure network stability.

(3) EV charging stations shall be provided with protection against the overload of input supply and output supply fittings.

(4) Where the charging station is to be connected on the distribution network for commercial purposes—

(a) an EV charging station operator shall notify the distribution network operator of their intention to set up such for effective planning; and

(b) the distribution network operator shall assess the impact the use of the charging station has on the wider electricity system.

(5) The electric vehicle parking place shall be such that the connection on the vehicle when parked for charging shall be within five meters from the electric vehicle charging point.

(6) Where the connection point is installed outdoors, or in a damp location, the equipment shall have an ingress protection code at least IP65 as defined in IEC standard 60529.

(7) Hazardous live parts shall not be exposed (this is to protect persons against electric shock).

(8) Every part of the electric apparatus shall be securely guarded and fenced unless they are safe by position or construction.

(9) The electric vehicle charging station shall be equipped with a protective device against the uncontrolled reverse power flow from the vehicle.

(10) The electric vehicle charging station shall have protection to prevent overvoltage or overloading of the battery.

(11) The electric vehicle charging point shall not be energised until it is connected to the vehicle.

(12) Power supply cables used in charging stations or charging points shall conform to the manufacturer's instructions or specifications or IEC 62893-1 standard.

(13) The safety provisions of all A.C. charging stations shall be in accordance with IEC 61851-1, IEC 61851-21 and IEC 61851-22.

(14) The safety provisions of all DC charging stations shall be in accordance with IEC 61851-1, IEC 61851-21, IEC 61851-23 and IEC 61851-24.

(15) All charging stations shall have the insulation resistance value as per the manufacturer's instructions or specifications or as stipulated in the relevant IEC 61851-1 standard.

(16) Electric vehicle supply equipment (EVSE) shall not create faults (typically through a circuit breaker or overcurrent protection), harmonics, and frequency imbalance in the distribution network.

(17) Any person who contravenes this section shall be guilty of an offence and liable to a fine not exceeding level 5 or to imprisonment for a period not exceeding six months or to both such fine and such imprisonment.

Periodic assessment of electric vehicle charging stations

7. (1) Every EV charging station owned, operated or managed by any operator must be subject to periodic assessment commencing on or before the first anniversary of connection of that charging station to a supply of electricity.

(2) Subsequent periodic assessments must take place at intervals of no more than twelve months.

(3) An EV charging station or EVSE must immediately be taken out of service by the electric vehicle charging station operator if the periodic assessment has not been carried out within twelve months from the date of first installation, or within twelve months from the date of the last inspection for that EV charging station.

(4) An EV charging station must be taken out of service immediately by the electric vehicle charging station operator if found to be unsafe at any time by any person, or in the event of an electric shock that results in a person being admitted to hospital.

(5) An EV charging station taken out of service in accordance with this section cannot be reconnected until its safety has been confirmed, following repair, and a new periodic assessment issued.

(6) An electric vehicle charging station operator must—

(a) establish and implement a safety assessment programme (periodic assessment) for regularly assessing the electrical safety of EVSE, conductors and fittings; and

(b) keep records of—

(i) the results of every periodic assessment; and

(ii) details of any issues found during the assessment; and

(iii) any actions required to be taken in relation to those issues; and

(c) retain a copy of these records, whether in hard copy or electronically, for at least five years; and

(d) on request, provide a copy of the records to the Authority within ten working days after the request.

(7) EVSE shall be capable of providing a technical basis for billing options, metering accuracy, and network connectivity.

(8) EVSE shall be capable of upgrades to enable smart-grid-capability through Open Charge Point Protocol (OCPP) transmission.

(9) Any person who contravenes this section shall be guilty of an offence and liable to a fine not exceeding level 5 or to imprisonment

for a period not exceeding six months or to both such fine and such imprisonment.

Electrical safety for users, equipment, and infrastructure

8. (1) No person may charge an electric vehicle by any, or any combination, of the following means—

(a) use of any EV plug adaptor that is not designed and approved by the EVSE manufacturer; or

(b) use of any EV adaptor that is not designed and approved by the car manufacturer and approved by the EVSE manufacturer; or

(c) cascading of two or more supply leads; or

(d) use of an extension lead; or

(e) use of a socket-outlet for the supply of more than one vehicle at a time; or

(f) use of an EVSE that is originally labelled by the manufacturer indicating it is not compatible with the 230V or 380V and 50Hz; or

(g) use of two or more connecting points to charge the same EV.

(2) No person may supply electricity for EV charging of an electric vehicle other than through a charging station.

(3) Any person who contravenes this section shall be guilty of an offence and liable to a fine not exceeding level 5 or to imprisonment for a period not exceeding six months or to both such fine and such imprisonment.

Electricity sources integration

9. (1) Where more than two electricity sources are interfaced for the purpose of supplying electricity to the charging station, the electricity sources shall be interfaced through a changeover switch.

(2) The changeover switch shall indicate the electricity source that will be powering at any given time.

(3) Any person who contravenes this section shall be guilty of an offence and liable to a fine not exceeding level 5 or to imprisonment for a period not exceeding six months or to both such fine and such imprisonment.

Conformity requirement

10. (1) Any person who imports or manufactures charging station equipment shall conform with the following standards—

(a) IEC 61851-1:2010 Electric vehicle conductive charging system – Part 1: General Requirements, for IC-CPD only, and only until 1 December 2018;

(b) IEC 61851:2017 Electric vehicle conductive charging system;

(c) IEC 61980-1:2015 Electric vehicle wireless power (WPT) systems – Part 1: General requirements;

(d) IEC 62196:2022, Plugs, socket-outlets, vehicle connectors and vehicle inlets – Conductive charging of electric vehicles;

(e) IEC 62262:2002 Degrees of protection provided by enclosures for electrical equipment against external mechanical impacts (IK code) IEC 62423;

(f) IEC 62752:2024 In-cable control and protection device for mode 2 charging of electric road vehicles (IC-CPD);

(g) IEC 62909-2:2019 Bi-directional grid-connected power converters - Part 2: Interface of GCPC and distributed energy resources;

(h) ISO/IEC 15118 Communication protocol between an electric vehicle (EV) and the charging station (EVSE) to enable smart, interoperable, and convenient charging;

(i) IEC 61000 electromagnetic compatibility (EMC) that ensures electronic equipment does not interfere with other devices and functions correctly in its electromagnetic environment.

(2) Any person who contravenes this section shall be guilty of an offence and liable to a fine not exceeding level 5 or to imprisonment

for a period not exceeding six months or to both such fine and such imprisonment.

Signage and information

11. (1) EV charging stations, other than in residential use, shall have posted signage, allowing only EVs to park in such charging spaces.

(2) EV charging stations shall have the following information affixed—

(a) outlet options;

(b) Voltage and amperage levels;

(c) time of use and fees; and

(d) safety information.

Power of enforcement

12. The Authority shall have the power to obtain, remove and test any EV charging station to enforce compliance with these regulations.

SCHEDULE

STANDARDS

ISO 15118-20: Vehicle-to-grid (V2G) standard.

IEC 63110: Protocol for management of EV charging and discharging infrastructure.

IEC 63119: Information exchange charging EVs everywhere – roaming.